06. Let S_{60} be the set of all divisors of 60 and D be the divisibility relation, aDb if and only if a divides b. Prove that (S_{60}, D) is a lattice. Check whether (S_{60}, D) is a (a) complete lattice (b) bounded lattice (c) complemented lattice, find the complement of each element.

[10] (CO4/K3)

07. (a) If $a, b \in S = \{1, 2, 3, 6\}$ and $a + b = \text{LCM}(a, b)$, $a.b = \text{GCD}(a, b)$ and $a' = \frac{6}{a}$, show that $\{S, +, .', 1, 6\}$ is a Boolean algebra. (5 Marks)

(b) A logistics company is planning delivery routes in two different cities A and B, both having 5 terminals. The streets of city A and B are given by

$\{u_1u_2, u_1u_4, u_1u_5, u_2u_3, u_2u_5, u_3u_4, u_3u_5, u_4u_5\}$, $\{v_1v_2, v_1v_3, v_1v_5, v_2v_3, v_2v_4, v_2v_5, v_3v_4, v_4v_5\}$ respectively.

(i) The company claims that the two city networks are structurally identical. Verify this by checking graph isomorphism and provide the mapping if true. (3 Marks)

(ii) Is it possible to traverse each street exactly once for inspection and return to the starting terminal? Justify your answer. (2 Marks)

[10] (CO4,5/K3)

08. An electrical engineer is designing a small communication circuit that connects four terminals: A: Power Source, B: Resistor, C: Capacitor, and D: Inductor. Each terminal must be directly connected to every other terminal using conducting wires to ensure complete signal flow.

(i) Model this network as a graph. (1 mark)

(ii) Is it possible to design this circuit such that no two wires are crossing each other in the plane. (2 marks)

(iii) Find the total number of walks of length 3 in this network? (4 marks)

(iv) List all Hamiltonian cycles starting from the power source. (3 marks)

[10] (CO5/K3)

09. A meteorological department has set up nine weather stations labelled A, B, C, D, E, F, G, H, I in a hierarchical network that forms a tree. Station A is the main root station, with B and C directly connected to it. Stations D and E are connected to B, while F and G are connected to C. Additionally, Station H is connected to E, and Station I is connected to G.

(a) Construct this as a tree. (2 Marks)

(b) Determine the traversal orders of the stations: list the stations in preorder, in order and post order. (3 Marks)

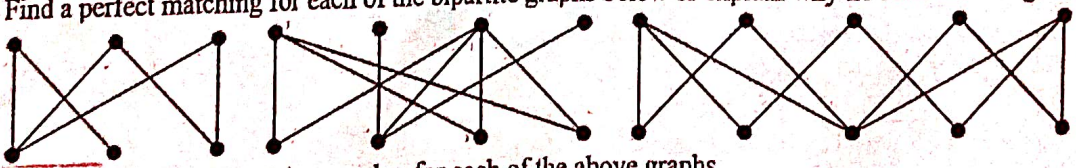
(c) Find the Eccentricity of all vertices, diameter, centre and radius of the constructed tree. (5 Marks)

[10] (CO5/K3)

10. A university plans to connect five Wi-Fi routers (A–E) at buildings $A(10, 20)$, $B(30, 10)$, $C(25, 35)$, $D(50, 25)$, and $E(45, 5)$. The distances between them determine the cable length to connect. Determine an acyclic network connecting all routers with minimum total cable length using minimum spanning tree.

[10] (CO5/K3)

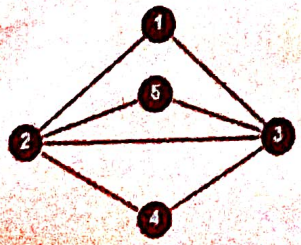
11. Find a perfect matching for each of the bipartite graphs below or explain why no such matching exists.



Also find the vertex covering number for each of the above graphs

[10] (CO5/K2)

12. Find the chromatic number and polynomial of the following graph.



[10] (CO5/K3)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)

VIT

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg No 24BAI1621

Final Assessment Test(FAT) - NOV/DEC 2025

Programme	B.Tech.	Semester	Fall Semester 2025-26
Course Code	BMAT205L	Faculty Name	Prof. Amit Kumar Rahul
Course Title	Discrete Mathematics and Graph Theory	Slot	D1+TD1+TDD1
		Class Nbr	CH2025260100769
Time	3 hours	Max. Marks	100

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: Learn proof techniques and concepts of inference theory
CO2: Use algebraic structures in applications
CO3: Counting techniques in engineering problems
CO4: Use Lattice and Boolean algebra properties in Digital circuits
CO5: Solve Science and Engineering problems using Graph Theory

Answer any 10 Questions (10 × 10 Marks)

01. (a) Determine whether the proposition

$$(P \vee Q) \wedge (\neg P \vee Q) \wedge (P \vee \neg Q)$$

is a tautology, contradiction, or contingency. **(5 Marks)**

(b) In a college of 250 students, 188 students are members of the drama club, 100 students are members of the music club, and 35 students are members of the arts club. Further 88 students are members of both drama and music clubs, 23 students are members of both music and art clubs, and 29 students are members of both drama and art clubs. If 19 students are members of all three clubs, how many students are not members of any club? **(5 Marks)**

[10] (CO1,3/K2)

02. Construct an argument to show that the following premises imply the conclusion "The keynote speaker did not cancel". "If the keynote speaker cancelled or the venue booking failed, then the conference will be re-scheduled and the proceedings will be delayed." "If the conference is rescheduled, then registration fees are refunded." "Registration fees were not refunded."

[10] (CO1/K3)

03. Construct the decoding table for the group code whose parity check matrix is

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Decode the following received words using the decoding table obtained. Among 011010, 101110, 111111., which cannot be decoded uniquely?

[10] (CO2/K2)

04. a) Let $G = \{x \in \mathbb{Z} : gcd(x, 12) = 1\}$. Prove that G is a group under multiplication modulo 12. Find the order of every element of G. **(6 marks)**

b)Let $G_1 = \mathbb{R} - \{\frac{1}{2}\}$ be a set with the binary operation $*$ defined on it by $a * b = a + b - 2ab$. Let $G_2 = \mathbb{R} - \{\frac{1}{2}\}$ be the group under the operation $\times$ (ie) usual multiplication. Prove that the function $f : (G_1, *) \to (G_2, \times)$ defined by $f(a) = 1 - 2a$ is a homomorphism. **(4 marks)**

[10] (CO2/K3)

05. Use the method of generating function to solve the following recurrence relation $L_n = L_{n-1} + L_{n-2}$, $n \geq 2$, $L_0 = 2$, $L_1 = 1$. (Lucas Sequence)

[10] (CO3/K2)